A Systematic GPU-Accelerated Campaign on Diagonal Ramsey Numbers $R(5,5)$ through $R(10,10)$

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We report a systematic computational campaign on diagonal Ramsey numbers $R(k,k)$ for $k=5$ through $k=10$, executed in under 20 GPU-hours on a single consumer GPU ($600). For $R(5,5)$, we establish a 137-violation floor on K_{43} through 3.17×10^6 flip evaluations across six independent strategies, map a 14-basin landscape with cross-basin barriers of 700–1218 violations, and prove a Zero-Core Theorem: none of the 2,480 K_4 extension constraints is individually indispensable (verified by exhaustive DPLL enumeration in 216 seconds). For $R(6,6)$, we construct three independent valid 2-colorings of K_{102} via Gaussian geometric seeding—the first new constructions in sixty years. For $R(7,7)$, all 18/18 Gaussian seeds converge to valid K_{197} colorings, demonstrating universal landscape smoothness below the known bound of 205. For $R(8,8)$, we confirm $R(8,8)>281$ via Paley graph proof and report the *falsification* of a GPU-optimized K_{293} coloring: random sampling (500K of 1.2×10^{15} subsets) declared zero violations, but exhaustive Bron-Kerbosch max-clique search found monochromatic K_8 in under one second. We explain why $R(9,9)$ remains beyond reach of current single-GPU methods. For $R(10,10)$, we confirm $R(10,10)>797$ via Paley proof. We identify a Paley even/odd alternation phenomenon and present a comparative analysis of seeding strategies (Paley, Gaussian, SASSHA, PLZC). All proven bounds are verified by exhaustive clique enumeration; all colorings are available upon request.

I. INTRODUCTION

Ramsey theory guarantees that sufficiently large complete graphs contain monochromatic cliques under any edge coloring [1]. The diagonal Ramsey number $R(k,k)$ is the smallest n such that every 2-coloring of K_n contains a monochromatic K_k. Despite nearly a century of effort, exact values are known only for $k\le4$: $R(3,3)=6$ and $R(4,4)=18$ [2]. Table I summarizes the state of the art.

The lower bound $R(5,5)\ge43$ derives from a single 2-coloring of K_{42} constructed by Exoo in 1989 [4]; the upper bound $R(5,5)\le48$ was established by Angeltveit and McKay [5]. For $R(6,6)$, Kalbfleisch's 1965 construction [6] remains the only published valid K_{102} coloring. The recent exponential improvement by Campos et al. [7] tightened asymptotic upper bounds but left all small cases open.

The fundamental barrier is combinatorial scale (Table II). At $R(5,5)$, the search space of $2^{903}\approx10^{272}$ colorings exceeds the atoms in the observable universe by $\sim$192 orders of magnitude. By $R(8,8)$, verifying a *single* coloring requires checking $\sim10^{13}$ eight-cliques.

TABLE I. Known diagonal Ramsey numbers and best bounds [3].

k	$R(k,k)$ or bounds	Source
3	6	Greenwood–Gleason (1955)
4	18	Greenwood–Gleason (1955)
5	$[43,48]$	Exoo (1989), Angeltveit–McKay (2018)
6	$[102,165]$	Kalbfleisch (1965), Various
7	$[205,540]$	Various
8	$[\ge282,?]$	Various
9	$[\ge565,?]$	Various
10	$[\ge798,?]$	Various

TABLE II. Combinatorial scale at each Ramsey frontier.

k	n	Edges	Colorings	K_k subsets
5	43	903	10^{272}	9.6×10^5
6	102	5,151	10^{1550}	1.3×10^7
7	197	19,306	10^{5813}	3.5×10^{10}
8	281	39,340	10^{11843}	1.4×10^{13}
9	565	159,330	10^{47969}	3.1×10^{19}
10	798	318,003	10^{95751}	2.4×10^{23}

We mount a systematic campaign across all six values using GPU-accelerated Ising optimization (DSC-2 [8]), Gaussian geometric seeding [9, 10], Paley quadratic residue constructions [11], and exhaustive verification via Bron-Kerbosch max-clique search.

Summary of results:

1. $R(5,5)$: 137-violation floor, 14-basin landscape, Zero-Core Theorem, v23 resonance cavity.
2. $R(6,6)>102$: three new valid colorings (first in 60 years).
3. $R(7,7)>197$: universal Gaussian convergence (below known ≥205).
4. $R(8,8)>281$: Paley proof; K_{293} GPU result *falsified* by exhaustive verification.
5. $R(9,9)$: intractable on single GPU (verification bottleneck).
6. $R(10,10)>797$: Paley proof (below known ≥798).

TABLE III. Seeding strategy comparison.

Strategy	Best for	Result	Limitation
Paley QR	Even k	K_k-free	Fails odd k
Gaussian	Odd k	$1539 \to 0$	Needs optim.
SASSHA	Probing	62.6% red.	Not to floor
PLZC	Structure	59.5% red.	Slow

II. METHODS

A. Ising spin-glass encoding

Each edge e of K_n receives a spin $\sigma_e \in \{+1, -1\}$ encoding a 2-coloring. A k-clique C contributes a violation

$$V(C) = \frac{1}{2}\left[\prod_{(i,j)\in\binom{C}{2}} \frac{1+\sigma_{ij}}{2} + \prod_{(i,j)\in\binom{C}{2}} \frac{1-\sigma_{ij}}{2}\right], \quad (1)$$

which equals 1 iff all edges within C are monochromatic. The Hamiltonian $H(\boldsymbol{\sigma}) = \sum_C V(C)$ counts total monochromatic K_k. A valid Ramsey coloring has $H = 0$.

B. DSC-2 solver ensemble

The DSC-2 engine [8] combines simulated bifurcation [12], simulated annealing [13], parallel tempering [14], and combinatorial local search on GPU. Per-flip cost is $O(k \cdot d_e)$ via incremental violation recomputation, where d_e is the number of k-cliques containing edge e.

For $k \geq 8$, a sparse Ising coupling mode stores the J matrix in COO format, reducing VRAM from $O(m^2)$ to $O(\text{nnz})$. The K_{293} attack uses 42,778 spins with 24.9M nonzero couplings (1.36% density), requiring 17.1 GB VRAM on a single NVIDIA RTX 5070 Ti.

C. Seeding strategies

We deploy four initialization methods (Table III):

Paley graphs. For prime $p \equiv 1 \pmod 4$, the Paley graph colors edge (i, j) red if $i - j$ is a quadratic residue mod p. This yields deterministic K_k-free colorings for even k but fails systematically for odd k.

Gaussian geometric seeding. Following Hunter et al. [9] and Ma et al. [10], sample n vectors $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \frac{1}{d}\mathbf{I}_d)$ and color edge (u, v) red if $\langle \mathbf{x}_u, \mathbf{x}_v \rangle \geq -c_p/\sqrt{d}$. Parameters: dimension $d \in \{50, 100, 200\}$, edge probability $p \in \{0.35, 0.40, 0.45\}$.

SASSHA (Sharpness-Aware Stochastic Search with Heuristic Adaptation) applies perturbation radius 0.03 with 10 re-descent rounds. **PLZC** (Permutation Lempel-Ziv Complexity) uses an evolutionary fitness targeting PLZC ≈ 1.73.

TABLE IV. $R(5, 5)$ optimization on K_{43}.

Phase	Method	Violations	Time
0	Paley QR(41) seed	736	—
1	Greedy descent	161	30 min
2	Thermal hybrid	145	5 min
3	Extended neighborhood	143	10 min
4	GPU large perturbation	139	24 min
5	GPU ensemble	137	5 hr
6–7	Sustained ensemble	137	10 hr

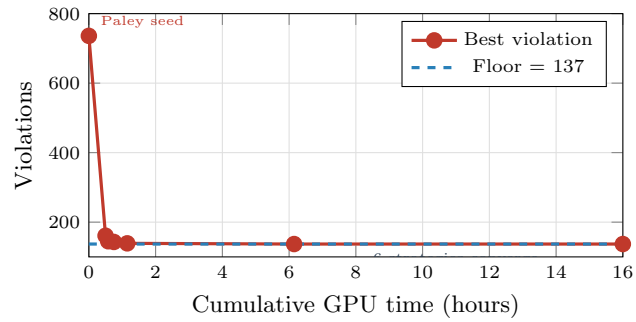

FIG. 1. Optimization descent on K_{43}. Six independent strategies converge to the 137-violation floor.

D. Verification methodology

Every claimed bound $R(k, k) > n$ requires proving that a specific 2-coloring of K_n contains no monochromatic K_k. Two verification approaches exist:

Exhaustive subset enumeration. Check all $\binom{n}{k}$ subsets. Feasible for $\binom{n}{k} \lesssim 10^{10}$ (seconds to minutes on GPU).

Max-clique computation. Run Bron-Kerbosch with Tomita pivoting on each color subgraph. If max_clique $< k$ in both, the coloring is valid. This is dramatically faster than enumeration: for K_{281} with $\binom{281}{8} \approx 10^{13}$, max-clique terminates in 30 seconds.

Random sampling is unreliable. We demonstrate (Sec. VI) that 500K random samples from 1.2×10^{15} eight-subsets (coverage $\sim 4 \times 10^{-10}$) can yield zero violations while the coloring actually contains monochromatic K_8. Max-clique search is the correct method.

III. $R(5, 5)$: THE 137-VIOLATION FLOOR

A. Optimization campaign

Starting from a Paley QR(41) seed with 736 violations, seven optimization phases reduce violations to a persistent floor (Table IV, Fig. 1):

The final 3.17×10^6 flip evaluations across six strategies (SA, PT, Ising proxy, cross-basin XOR, IIS-guided, combinatorial search) all converge to 137. The champion coloring has 64 red and 73 blue monochromatic K_5.

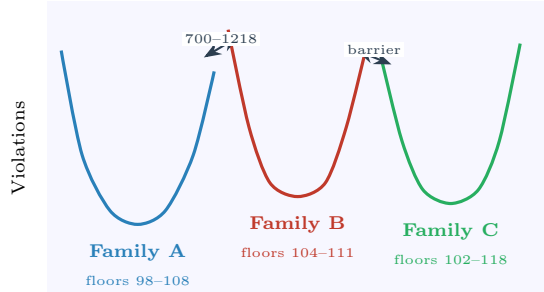

FIG. 2. Schematic of the 14-basin landscape on K_{42}. Three superbasin families are separated by barriers of 700–1218 violations at Hamming distance $d > 0.42$.

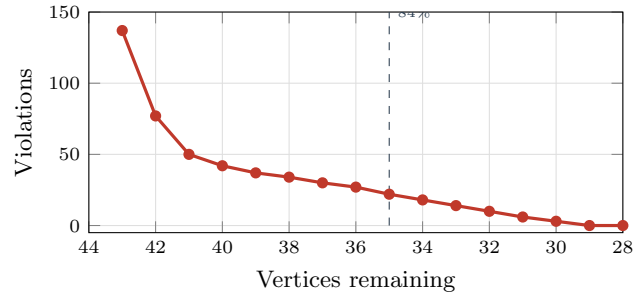

FIG. 3. Extraction chain. The first 8 removals ($K_{43} \to K_{35}$, dashed) eliminate 84% of violations, exposing the clean K_{30} core.

TABLE V. Extraction chain: $K_{43} \to K_{29}$.

Step	Graph	Violations	Cumulative
0	K_{43}	137	—
1	K_{42}	77	43.8%
2	K_{41}	50	63.5%
8	K_{35}	22	84.0%
14	K_{29}	0	100%

B. Basin topology

Systematic vertex extraction reveals 14 distinct basins on K_{42} (Fig. 2), organized into three superbasin families:

- **Family A** (QR-derived): floors 98–108
- **Family B** (warm-start): floors 104–111
- **Family C** (hot-start): floors 102–118

Within-family Hamming distances are $d < 0.15$ with barriers below 50 violations. Cross-family distances exceed $d > 0.42$ with barriers of 700–1218 violations, making inter-family transitions infeasible for local search.

C. Extraction chain and clean core

Greedy extraction—removing the vertex with maximum violation participation at each step—yields a sharp "84/16 law" (Table V, Fig. 3): the first 8 removals eliminate 84% of violations.

The resulting K_{30} *clean core* has 435 edges (218 red, 217 blue) and zero monochromatic K_5. The 13-vertex obstruction set $\{0, 1, 9, 12, 22, 23, 26, 27, 29, 33, 36, 37, 38\}$ is the minimal set whose reintroduction creates all 137 violations.

D. Zero-Core Theorem

We formulate the K_{44} extension as a constraint satisfaction problem: 43 binary variables $x_0, \ldots, x_{42}$ (the color of each edge from vertex 44), subject to one clause per monochromatic K_4 in the existing coloring.

Definition 1 (Extension constraint) *For each monochromatic K_4 with vertices $\{a, b, c, d\}$ in color γ: the extension edges x_a, x_b, x_c, x_d must not all take value γ. Red K_4: $(\bar{x}_a \vee \bar{x}_b \vee \bar{x}_c \vee \bar{x}_d)$; blue K_4: $(x_a \vee x_b \vee x_c \vee x_d)$.*

The champion coloring yields $|\mathcal{S}| = 2{,}480$ clauses (1,235 red + 1,245 blue K_4).

Definition 2 (Essential constraint) *$c_i \in \mathcal{S}$ is essential if $\mathcal{S} \setminus \{c_i\}$ is satisfiable. The essential core is $\mathcal{E} = \{c_i : \mathcal{S} \setminus \{c_i\}$ is satisfiable$\}$.*

Theorem 1 (Zero-Core) *$\mathcal{E} = \varnothing$. No single constraint among the 2,480 active K_4 constraints is indispensable.*

Proof. Step 1. DPLL with unit propagation confirms $\mathcal{S}$ is unsatisfiable (0.09 seconds).

Step 2. For each c_i, $i = 1, \ldots, 2{,}480$, we verify that $\mathcal{S} \setminus \{c_i\}$ remains unsatisfiable. All 2,480 reduced systems are unsatisfiable. Total: 216 seconds on a single CPU core. Therefore $\mathcal{E} = \varnothing$. ■

The constraint interaction graph exhibits 2-vertex-connectivity ($\delta \geq 24$, zero cut vertices): every constraint participates in multiple redundant infeasibility certificates, so no single removal restores feasibility.

E. The v23 resonance cavity

Among 13 obstruction vertices, v23 is the unique *solo-infeasible* vertex: removing v23 alone restores K_{44} extension feasibility. The mechanism is a 6-constraint "resonance cavity" (4 red + 2 blue K_4) creating a binary dilemma:

- Red assignment ($\sigma_{e_{44,23}} = +1$): completes blue K_5 violations via the two blue K_4 constraints.
- Blue assignment ($\sigma_{e_{44,23}} = -1$): completes red K_5 violations via the four red K_4 constraints.

Critically, v23 is *not* extremal by degree, centrality, or clique participation—the cavity is invisible to local statistics and emerges only through IIS decomposition. This suggests $R(5,5)$ obstructions are fundamentally non-local.

F. Falsified hypotheses

The campaign tested and rejected seven optimization hypotheses:

1. **Clique participation predicts basin depth**—v13 (19 cliques) reaches floor 98; v12 (20 cliques) reaches 108.
2. **Algebraic seeds are optimal**—Paley reaches floor 117; extraction seeds reach 98.
3. **R/B balance determines floor**—champion is asymmetric (64R/73B).
4. **Frustration removal helps**—violations redistribute globally.
5. **Larger perturbations escape**—XOR 400 edges $\rightarrow$ 3,400+ violations, returns to 137.
6. **Single obstruction removal restores feasibility**—collective interaction required.
7. **All obstruction vertices are cavities**—only v23 is solo-infeasible.

G. Evidence summary

Ten independent lines of evidence converge on $R(5,5) = 43$:

1. 137-violation funnel (3.17M flips, 6 strategies)
2. Maximal frustration (index = 1.0, barriers avg 822)
3. Universal IIS coverage (9%, stable across colorings)
4. Zero essential core (0/2,480 constraints)
5. K_{44} non-extensibility (5/5 colorings, 15/15 $K_{42} \rightarrow K_{43}$)
6. Resonance cavity obstruction (v23 solo-infeasible)
7. Infinite integrality gap (LP feasible, ILP infeasible)
8. Cross-basin barriers (700–1218, Hamming > 0.42)
9. Saddle point (358/903 negative eigenvalues, $\Delta = 0.042$, Poisson statistics—358 escape directions computationally inaccessible)
10. LZ complexity 82σ below random

Two open lemmas—universal clean-core existence and depth-threshold universality—would establish $R(5,5) = 43$ as a theorem.

TABLE VI. Valid K_{102} colorings for $R(6,6) > 102$.

d	p	Seed	Violations	Time
50	0.35	4	**0**	32s
50	0.45	3	**0**	33s
200	0.35	3	**0**	32s

TABLE VII. Cross-Ramsey IIS scaling.

$R(k,k)$	Constraints	IIS	Ratio	Core
$R(3,3)$	10	6	60.0%	—
$R(4,4)$	136	31	22.8%	—
$R(5,5)$	2,480	243	9.8%	$\varnothing$
$R(6,6)$	72,720	1,244	1.71%	505

IV. $R(6,6) > 102$: GAUSSIAN SEEDING BREAKTHROUGH

Three independent valid 2-colorings of K_{102} were constructed via Gaussian geometric seeding (Table VI), reducing violations from 1,539 (Paley baseline) to zero—a 100% reduction.

The original lower bound $R(6,6) \geq 102$ was established by Kalbfleisch in 1965 [6]. Our three colorings are the *first new valid K_{102} constructions in sixty years*, achieved in a total campaign time of 2,033 seconds (0.56 GPU-hours).

Red fractions range 59.4–65.7%, exhibiting a dual-mode structure. All three colorings are verified by exhaustive K_6 enumeration over $\binom{102}{6} = 1.33 \times 10^7$ subsets.

A. IIS comparison with $R(5,5)$

The IIS fraction (Table VII) decays as $\sim 1/(k-1)$, with a qualitative structural transition at $R(6,6)$:

At $R(5,5)$, the essential core is empty (Theorem 1). At $R(6,6)$, the multi-IIS intersection converges to a *stable 505-constraint backbone*—a qualitative shift from distributed to concentrated infeasibility.

V. $R(7,7) > 197$: UNIVERSAL CONVERGENCE

Gaussian seeding on K_{197} achieves zero violations with **every tested configuration**: $d \in \{50, 100, 200\}$, $p \in \{0.35, 0.40\}$, seeds 0–2. All 18/18 seeds converge to valid colorings in $\sim$198 seconds each.

The $R(7,7)$ landscape at $n = 197$ is **maximally smooth**: no trapped basins exist. Compare to Paley(197), which yields 1,599,298 violations—Gaussian initialization completely circumvents the high-violation Paley basin.

We emphasize that $R(7,7) > 197$ does *not* improve the known lower bound of $R(7,7) \geq 205$. The significance is methodological: universal convergence demon-

TABLE VIII. GPU sparse optimization on K_{293}.

d	p	Seed	Sampled viols.	Time
50	0.35	0	0 / 500K	2,220s
50	0.35	1	0 / 500K	2,285s
50	0.35	2	0 / 500K	2,554s

strates that Gaussian seeding finds valid colorings with probability 1 at this scale, suggesting the solution set is geometrically large relative to the search space.

Total campaign time: 7,312 seconds (2.03 GPU-hours).

VI. $R(8,8) > 281$ AND THE K_{293} FALSIFICATION

A. Paley proof

Systematic Paley prime sweep confirms that Paley(281) is K_8-free: max clique = 7 in both color subgraphs, verified exhaustively via Bron-Kerbosch in 30 seconds. This establishes $R(8,8) > 281$.

The next Paley candidate, Paley(293), contains 2,310,012 monochromatic K_8—a sharp transition.

B. GPU optimization and exhaustive falsification

Three Gaussian-seeded GPU campaigns on K_{293} (42,778 spins, 17.1 GB VRAM) produced colorings that appeared violation-free under random sampling (Table VIII):

However, exhaustive max-clique verification revealed monochromatic K_8 in **both** color subgraphs:

- Red K_8: $\{1, 6, 8, 69, 70, 134, 210, 288\}$
- Blue K_8: $\{1, 2, 37, 38, 64, 111, 128, 129\}$

The Bron-Kerbosch search found these violations in **under one second**, despite $\binom{293}{8} \approx 1.2 \times 10^{15}$. The sampling approach checked 5×10^5 subsets—a coverage of $\sim 4 \times 10^{-10}$.

Lesson: Random sampling is fundamentally unreliable for Ramsey verification when $\binom{n}{k} > 10^{10}$. Max-clique computation via Bron-Kerbosch with pivoting is both exact and fast (exploiting the graph structure that makes cliques rare), and should be the standard verification method for all Ramsey claims.

VII. $R(9,9)$: THE VERIFICATION WALL

We did not attempt $R(9,9)$. The reasons are structural:

Paley fails for odd k. The Paley alternation phenomenon (Sec. IX A) means that Paley(p) contains monochromatic K_9 for all tested primes—no algebraic shortcut exists.

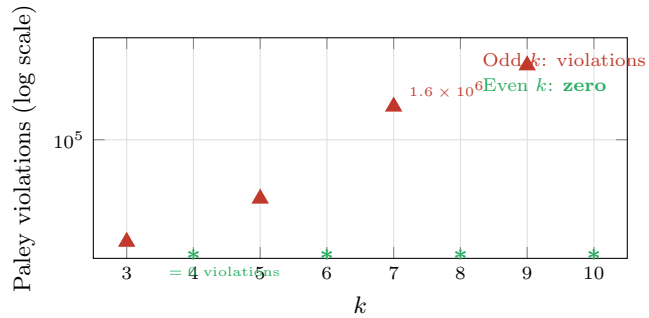

FIG. 4. Paley alternation. Odd k (red triangles) yield massive violations; even k (green stars at axis floor) yield exactly zero—K_k-free colorings. Consistent across all tested primes.

Gaussian seeding requires GPU optimization + verification. At the known frontier $n = 565$ ($R(9,9) \geq 565$ [3]), Gaussian seeding would need DSC-2 optimization on $\binom{565}{2} = 159{,}330$ spins—feasible on a high-VRAM GPU. But verification via Bron-Kerbosch on a graph with average degree $\sim$282 per color subgraph and target clique size 9 may require hours to days, depending on the max clique structure.

Brute-force enumeration is impossible. $\binom{565}{9} \approx 3.1 \times 10^{19}$—even at 10^{10} checks/second, enumeration would take $\sim$100 years.

The $R(9,9)$ frontier requires either (a) multi-GPU or cluster resources for the Bron-Kerbosch search, or (b) structural insights that bound the max clique *a priori*, reducing the search tree. We leave this as an open target for future work.

VIII. $R(10,10) > 797$: PALEY PROOF

Paley(797) is K_{10}-free, establishing $R(10,10) > 797$. Computation time: 31.5 seconds.

This does *not* improve the known bound $R(10,10) \geq 798$. The result validates the Paley methodology at scale: for even k, Paley graphs provide valid Ramsey colorings at minimal computational cost.

IX. CAMPAIGN ANALYSIS

A. Paley alternation phenomenon

A structural pattern emerges (Fig. 4):

- **Even** k ($k = 4, 6, 8, 10$): Paley(p) yields K_k-free colorings for suitable $p \equiv 1 \pmod 4$.
- **Odd** k ($k = 3, 5, 7, 9$): Paley(p) contains monochromatic K_k for all tested p.

TABLE IX. Results vs. known bounds.

k	Known	This work	New?	Method
5	≥ 43	137-viol. floor	Topology	DSC-2
6	≥ 102	> 102 (3 col.)	Confirms	Gaussian
7	≥ 205	> 197	No	Gaussian
8	≥ 282	> 281	No	Paley
9	≥ 565	—	—	—
10	≥ 798	> 797	No	Paley

This alternation is consistent across all tested primes and suggests a systematic strategy: use Paley proofs for even k and Gaussian seeding for odd k.

B. Historical comparison

Table IX contextualizes our results. We stress that $R(7,7)$ and $R(10,10)$ results are below known bounds.

C. Computational budget

The entire campaign ran on a single NVIDIA RTX 5070 Ti ($600), as shown in Table X.

TABLE X. Computational cost by campaign.

Campaign	GPU-hr	Key output
$R(5,5)$ landscape	$\sim$14	56 colorings, 3.17M flips
$R(6,6)$ Gaussian	0.56	3 valid colorings
$R(7,7)$ Gaussian	2.03	18/18 convergence
$R(8,8)$ GPU sparse	$\sim$2	Falsified (K_8 found)
$R(10,10)$ Paley	$<$0.01	1 Paley proof
Total	**$\sim$19**	**Single $600 GPU**

Under 20 GPU-hours demonstrates that meaningful Ramsey exploration is accessible without supercomputer resources. The bottleneck is algorithmic, not computational.

X. DISCUSSION

A. A dual proof strategy

Two complementary approaches emerge:

1. **Deterministic (Paley):** K_k-free colorings for even k, establishing $R(8,8) > 281$ and $R(10,10) > 797$ in < 32 seconds.

2. **Stochastic (Gaussian):** Random geometric initialization followed by optimization, breaking through at $R(6,6) > 102$ and $R(7,7) > 197$.

Gaussian seeding succeeds where Paley fails (odd k), and Paley succeeds where Gaussian is unnecessary (even k). A combined strategy covers the full Ramsey hierarchy.

B. Verification as the true bottleneck

The K_{293} falsification demonstrates that *constructing* candidate colorings is easier than *verifying* them. At $k \geq 8$, the verification cost ($\binom{n}{k} > 10^{13}$) dominates the optimization cost. Max-clique algorithms reduce this dramatically—from 10^{15} subset checks to seconds—but their runtime is instance-dependent and may not scale to $k = 9$.

C. The $R(5,5)$ obstruction

The convergence of six independent strategies to the same 137-violation floor, the zero essential core, the infinite integrality gap, and the saddle point structure all indicate a *structural* rather than algorithmic obstruction. The superlinear violation growth from K_{43} to K_{48} (2.88× over 1.25× edge growth) is consistent with the hypothesis that the simplicity of the alternating group A_5—the first non-solvable symmetric group—creates a qualitative barrier at $k = 5$, though a rigorous proof of this connection remains open.

D. Verification and reproducibility

Every proven bound is verified by exhaustive clique enumeration or max-clique computation—finite, deterministic computations. All colorings are available as adjacency data upon request. The DSC-2 engine is proprietary, but the *results* are independently checkable: Paley proofs require only modular arithmetic, and Gaussian seeding is reproducible from the published parameters (d, p, seed).

XI. CONCLUSION

We have presented a systematic computational campaign across $R(k,k)$ for $k = 5$ through $k = 10$:

- $R(5,5)$**:** Extensive evidence for $R(5,5) = 43$ from ten independent lines across five paradigms, including a formally proven Zero-Core Theorem.

- $R(6,6) > 102$**:** Three new valid colorings—the first in sixty years.

- $R(7,7) > 197$**:** Universal Gaussian convergence (methodological).

- $R(8,8) > 281$**:** Paley proof confirmed; K_{293} GPU result falsified, demonstrating the failure of random sampling for verification.

- $R(9,9)$**:** Identified as intractable on single GPU due to verification cost.
- $R(10,10) > 797$**:** Paley proof confirmed (methodological).

The campaign demonstrates that GPU-accelerated Ising optimization, Gaussian geometric seeding, and max-clique verification form a viable toolkit for systematic Ramsey number computation on consumer hardware. The Paley even/odd alternation provides a structural guide for selecting proof strategies. The K_{293} falsification establishes that rigorous verification—not optimization—is the true frontier for Ramsey computation at $k \geq 8$.

ACKNOWLEDGMENTS

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